

R IGNITE 2026 • SESSION 2

Interactive R Primer with Real NHANES Health Data

Import from a URL • Explore • Test hypotheses • Fit logistic regression

A 100-minute hands-on workflow for epidemiology, nursing, pharmacy and health sciences

Where Session 2 takes us



Import

URL → data frame



Inspect

structure + missingness



Explore

summaries + plots



Infer

t-test • χ^2 • ANOVA



Model

logistic regression

Core principle: every statistical method starts with a health question and ends with an interpretation—not with R syntax.

Learning outcomes



- Import a real health dataset directly from a web URL
- Recognize numeric, categorical and binary variables
- Handle duplicates and missing observations deliberately
- Select and run a two-sample t-test, chi-square test and one-way ANOVA
- State H_0 and H_1 in scientific language
- Check key assumptions before interpreting output
- Fit a multivariable logistic regression model
- Translate coefficients into odds ratios and predicted probabilities

Meet the NHANES teaching dataset

What it is

A teaching CSV derived from NHANES data collected in 2009–2012. It contains 10,000 rows and many demographic, examination, health and lifestyle variables.

Why it works for this lab

One dataset supports continuous outcomes (BMI, blood pressure), categorical exposures (Gender, Education, PhysActive), and a binary outcome (Diabetes).

Note: this simplified lab does not perform survey-weighted population inference.

Import directly from the working URL

BASE R — NO PACKAGE REQUIRED

```
url <- "https://raw.githubusercontent.com/statOmics/PSLSDData/main/NHANES.csv"
NHANES <- read.csv(url)

dim(NHANES)
head(NHANES)
```



Why use a URL?

Reproducible: everyone reads the same source.
Portable: no local file path to fix.
Transparent: the source is explicit in the script.

What if the network fails?

Use the download data in my GitHub page. Upload as
`read.csv("NHANES.csv")`.

Verify the import before analysis

R CODE

```
dim(NHANES)
names(NHANES)
str(NHANES)
summary(NHANES$BMI)
table(NHANES$Diabetes, useNA = "ifany")
```

- Expected raw dimensions: 10,000 rows × 76 columns
- Confirm exact capitalization of variable names
- Look for NA values before choosing a method
- Ask: What is the unit of observation?
- Do not interpret a model until the data structure makes sense

Important data issue: repeated IDs

The teaching CSV contains repeated participant IDs. For this workshop we keep one row per ID before person-level analyses.

CREATE ADULT ANALYTIC DATA

```
sum(duplicated(NHANES$ID))
```

```
dat <- NHANES[!duplicated(NHANES$ID), ]  
dat <- subset(dat, Age >= 20)
```

```
dim(dat)
```

- This is a teaching simplification
- Always investigate why duplicates exist in real research data
- Define the analytic population before testing hypotheses

Variables we will use

Variable	Type	Meaning	Used for
BMI	continuous	kg/m ²	t-test
Gender	categorical	male / female	t-test, χ^2
Diabetes	binary	No / Yes	χ^2 , logistic
HealthGen	categorical	Excellent → Poor	ANOVA
BPSysAve	continuous	average systolic BP	ANOVA
Age	continuous	years	logistic
PhysActive	binary	No / Yes	logistic

Missingness is part of the analysis

R CODE

```
vars <- c("BMI", "Gender", "Diabetes",  
         "HealthGen", "BPSysAve",  
         "Age", "PhysActive")  
  
sapply(dat[vars], function(x) sum(is.na(x)))
```

- Different analyses may use different complete-case subsets
- Missingness can change sample size and interpretation
- Never silently replace NA with 0
- Record the analysis n when reporting results

Explore before you test

R CODE

```
summary(dat$BMI)
table(dat$Gender)
prop.table(table(dat$Diabetes))

with(dat, tapply(BMI, Gender, mean, na.rm=TRUE))
with(dat, tapply(BMI, Gender, sd, na.rm=TRUE))
```

- Describe the sample before making inferential claims
- Check distributions and group sizes
- Ask whether a difference is practically meaningful
- Plots often reveal problems that a p-value hides

Visualization: BMI by gender

R CODE

```
boxplot(BMI ~ Gender, data = dat,  
        xlab = "Gender", ylab = "BMI (kg/m²)",  
        main = "BMI by Gender")
```

Read the plot before running the test

- Compare medians and spread
- Look for extreme observations
- Notice overlap between groups
- Do not infer “significance” from the boxplot alone

Hypothesis testing: a reusable framework

1. Question

What scientific comparison are we making?

2. Hypotheses

State H_0 and H_1 before seeing the p-value.

3. Assumptions

Are observations independent? Are model conditions plausible?

4. Evidence

Use test statistic, CI, p-value and effect size.

5. Interpretation

Translate output back into health-science language.

Test 1 — two-sample t-test

Research question

Among adults in this teaching dataset, does mean BMI differ between gender groups?

$H_0: \mu_{\text{male}} = \mu_{\text{female}}$

$H_1: \mu_{\text{male}} \neq \mu_{\text{female}}$

Why Welch's t-test?

The outcome BMI is continuous and Gender defines two independent groups. R's `t.test()` defaults to Welch's method, which does not assume equal group variances.

Run and interpret the t-test

R CODE

```
tt <- t.test(BMI ~ Gender, data = dat)
tt

with(dat, tapply(BMI, Gender, mean, na.rm=TRUE))
```

- Report both group means
- Report the estimated difference and 95% CI
- Use the p-value as evidence against H_0 —not as the probability H_0 is true
- Ask whether the magnitude matters clinically or scientifically

Interpretation template: “Mean BMI was [higher/lower/similar] in ___ than ___; the estimated difference was ___ (95% CI ___), $p = ___$.”

Test 2 — chi-square test of association

Research question

Is diabetes status associated with gender in the adult analytic sample?

H_0 : Gender and Diabetes are independent.
 H_1 : They are associated.

What the test uses

A contingency table of counts. The chi-square statistic compares observed cell counts with counts expected under independence.

Run the chi-square test

R CODE

```
tab <- with(dat, table(Gender, Diabetes, useNA="no"))
tab
prop.table(tab, margin = 1)

chi <- chisq.test(tab)
chi
chi$expected
```

- Inspect counts and row percentages
- Check expected counts before trusting the approximation
- A significant result shows association—not causation
- If expected counts are sparse, consider Fisher's exact test

Test 3 — one-way ANOVA

Research question: Does mean average systolic blood pressure (BPSysAve) differ across self-reported general-health categories?

H₀ All population group means are equal.

H₁ At least one group mean differs.

- Outcome: BPSysAve (continuous)
- Group: HealthGen (5 categories)
- ANOVA is an omnibus test
- A significant ANOVA does not identify which groups differ

Run ANOVA and post-hoc comparisons

R CODE

```
anova_dat <- subset(dat, !is.na(BPSysAve) & !is.na(HealthGen))
anova_dat$HealthGen <- factor(anova_dat$HealthGen)

fit_aov <- aov(BPSysAve ~ HealthGen, data=anova_dat)
summary(fit_aov)
TukeyHSD(fit_aov)
```

- Start with the omnibus F-test
- If warranted, use Tukey HSD for pairwise comparisons
- Report group means alongside p-values
- Remember: self-reported health and blood pressure are observational measures

ANOVA assumptions and diagnostics

R CODE

```
par(mfrow=c(1,2))  
plot(fit_aov, which=1) # residuals vs fitted  
plot(fit_aov, which=2) # normal Q-Q  
par(mfrow=c(1,1))
```

- Independent observations
- Approximately normal residuals within the model
- Comparable within-group variability
- Diagnostics are about model residuals—not merely whether the raw outcome is perfectly normal

Why logistic regression?

Outcome

Diabetes has two levels: No / Yes. Ordinary linear regression is not designed for a binary response.

Goal

Model the probability—or equivalently the odds—of diabetes as a function of several participant characteristics at the same time.

Teaching model: association, not clinical diagnosis or causal effect.

Prepare the logistic-regression data

R CODE

```
logit_dat <- subset(dat,  
  !is.na(Diabetes) & !is.na(Age) &  
  !is.na(BMI) & !is.na(Gender) &  
  !is.na(PhysActive))  
  
logit_dat$Diabetes01 <- ifelse(logit_dat$Diabetes == "Yes", 1, 0)  
logit_dat$Gender <- factor(logit_dat$Gender)  
logit_dat$PhysActive <- factor(logit_dat$PhysActive)
```

- Make the event coding explicit: Yes = 1
- Check reference levels with levels()
- Record complete-case sample size
- Coding decisions change coefficient interpretation

Fit the multivariable logistic model

R CODE

```
fit_logit <- glm(  
  Diabetes01 ~ Age + BMI + Gender + PhysActive,  
  data = logit_dat,  
  family = binomial  
)  
  
summary(fit_logit)
```

- Each coefficient is adjusted for the other predictors in the model
- Raw coefficients are on the log-odds scale
- The sign tells the direction of association
- Exponentiating coefficients gives odds ratios

From coefficients to odds ratios

R CODE

```
OR <- exp(coef(fit_logit))  
CI <- exp(confint.default(fit_logit))  
round(cbind(OR, CI), 2)
```

How to read an odds ratio

- $OR > 1$: higher odds of diabetes
- $OR < 1$: lower odds of diabetes
- $OR = 1$: no multiplicative change in odds
- For continuous predictors, the OR corresponds to a one-unit increase unless you rescale the variable

Predicted probabilities: easier to communicate

R CODE

```
new_people <- data.frame(  
  Age = c(30, 50, 70),  
  BMI = c(24, 30, 36),  
  Gender = factor(c("female", "female", "female"),  
    levels=levels(logit_dat$Gender)),  
  PhysActive = factor(c("Yes", "Yes", "No"),  
    levels=levels(logit_dat$PhysActive))  
)  
new_people$prob <- predict(fit_logit, newdata=new_people,  
  type="response")
```

- A probability is often more intuitive than log-odds
- Keep all other predictors explicit when comparing profiles
- These are model-based teaching estimates—not clinical risk scores

One dataset, four inferential questions

Method	Outcome	Design cue	Session example
t-test	Continuous	2 groups	Mean BMI by Gender
Chi-square	Categorical	2 categorical vars	Gender × Diabetes
ANOVA	Continuous	3+ groups	BPSysAve by HealthGen
Logistic regression	Binary	Multiple predictors	Diabetes ~ Age + BMI + Gender + PhysActive

Participant challenge



In pairs, choose one question and complete the full workflow.

- State the health question
- Identify outcome and explanatory variable(s)
- State H_0/H_1 when appropriate
- Run a descriptive summary and plot
- Choose the statistical method
- Check the key assumptions
- Report the main numerical result
- Write a 2–3 sentence health-science interpretation

Common mistakes to catch early

R / data mistakes

- Wrong capitalization in variable names
- Forgetting NA handling
- Treating categorical variables as numeric
- Ignoring duplicate IDs
- Changing the outcome coding without noticing

Statistical mistakes

- Choosing a test from the p-value you want
- Reporting significance without effect size
- Calling association causation
- Interpreting odds ratio as risk ratio
- Skipping assumptions

Session 2 takeaways

A strong analysis is a chain of defensible decisions.

DATA

Import → inspect → clean

QUESTION

Match variables to the
scientific question

METHOD

Choose test/model from
design + assumptions

INTERPRET

Effect + uncertainty +
context

Next: Session 3 asks what p-values, confidence intervals and probability actually mean—and introduces the Bayesian alternative.